LESSON 2.4

Represent and Describe Data in a Box Plot

Set B · Version A · Second form · supported practice

MASTERY CHECK

- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can make and read a box plot to summarize a data set.

WHY THIS MATTERS The basketball league wants to compare scoring across teams. Here are the points scored by each player on the Wildcats in their last game: 4, 6, 8, 10, 12, 14, 15, 18, 22, 24, 28. The league needs a quick visual that shows the spread, the middle 50%, and any high or low scorers. Time to build a box plot!

REMEMBER

Represent and Describe Data in a Box Plot

Box Plots & 5-Number Summary. Formula: IQR = Q3 − Q1 A box plot is a simple picture that shows how a set of data is spread out, using just five key numbers: the lowest value, the first quartile (Q1), the median (middle), the third quartile (Q3), and the highest value.

1. 5-Number Summary: Minimum, Lower Quartile (Q1), Median (Q2), Upper Quartile (Q3), Maximum.
2. Draw the Box: The box spans from Q1 to Q3 and contains the middle 50% of data.
3. Whiskers: Draw lines from the box out to the minimum and maximum

Words: **Box Plot *(Diagrama de caja)*** **Median *(Mediana)*** **Quartile *(Cuartil)*** **Interquartile Range *(Rango intercuartílico)*** **Data distribution *(Distribución de datos)***

1 **Circle the letter of the best answer. Show how you know.**

The values 4, 6, 9, 12 and 19 are already in order. What is the median?

(A) 6 (B) 9 (C) 10 (D) 12

Put the data in order _____

WORK

1. **Order** *Smallest to largest.*
2. **Count** *How many values?*
3. **Locate** *Middle, or add and divide.*
4. **Answer** *Say which measure it is.*

2 **Circle the letter of the best answer. Show how you know.**

Order these values from least to greatest: 6, 13, 4, 9 and 3. What is the median?

(A) 4
(B) 6
(C) 7
(D) 13

Put the data in order _____

WORK

1. **Order** *Smallest to largest.*
2. **Count** *How many values?*
3. **Locate** *Middle, or add and divide.*
4. **Answer** *Say which measure it is.*

3 **Circle the letter of the best answer. Show how you know.**

Find the median of 4, 8, 6, 10.

(A) 6
(B) 7
(C) 8
(D) 28

Put the data in order _____

WORK

1. **Order** *Smallest to largest.*
2. **Count** *How many values?*
3. **Locate** *Middle, or add and divide.*
4. **Answer** *Say which measure it is.*

4 **Circle the letter of the best answer. Show how you know.**

Why must you put the values in order before finding the median?

(A) The median is the middle value in order
(B) It makes them easier to add
(C) It changes the largest value
(D) Order does not matter

SHOW YOUR WORK

5 Circle the letter of the best answer. Show how you know.

Two basketball teams' scoring data are displayed as box plots. Team A: Min=40, Q1=52, Median=60, Q3=68, Max=75. Team B: Min=35, Q1=55, Median=61, Q3=65, Max=80. A fan says Team B is better because they scored 80 points once. What is Team A's IQR?

(A) 10
(B) 16
(C) 35
(D) 60

Put the data in order _____

WORK

1. **Order** *Smallest to largest.*
2. **Count** *How many values?*
3. **Locate** *Middle, or add and divide.*
4. **Answer** *Say which measure it is.*

6 Circle the letter of the best answer. Show how you know.

Two basketball teams' scoring data are displayed as box plots. Team A: Min=40, Q1=52, Median=60, Q3=68, Max=75. Team B: Min=35, Q1=55, Median=61, Q3=65, Max=80. A fan says Team B is better because they scored 80 points once. What is Team B's IQR?

(A) 10
(B) 16
(C) 45
(D) 61

Put the data in order _____

WORK

1. **Order** *Smallest to largest.*
2. **Count** *How many values?*
3. **Locate** *Middle, or add and divide.*
4. **Answer** *Say which measure it is.*

EXPLAIN YOUR THINKING

The IQR is Q3 − Q1 = 26 − 10 = 16. What does this tell you about the middle 50% of scorers?

SENTENCE STARTER The middle 50% of players scored between ___ and ___ points. The IQR of ___ means the middle half of the data is spread across ___ points. This tells me ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? □ 4 · I can teach it □ 3 · I've got it □ 2 · I need a hint □ 1 · I need help

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WHY THIS MATTERS The basketball league wants to compare scoring across teams. Here are the points scored by each player on the Wildcats in their last game: 4, 6, 8, 10, 12, 14, 15, 18, 22, 24, 28. The league needs a quick visual that shows the spread, the middle 50%, and any high or low scorers. Time to build a box plot!

PART 1 Understand the idea

models and meaning

1 **Write the letter of the matching item on each line.**

Match each team's ordered scores to its five-number summary (min, Q1, median, Q3, max).

_____ Hawks: 5, 8, 11, 15, 18, 21, 27

_____ Comets: 10, 13, 16, 20, 24, 28, 33

_____ Jets: 2, 6, 9, 14, 19, 23, 30

_____ Suns: 4, 7, 10, 16, 22, 25, 31

(A) Min 5, Q1 8, Median 15, Q3 21, Max 27

(B) Min 10, Q1 13, Median 20, Q3 28, Max 33

(C) Min 2, Q1 6, Median 14, Q3 23, Max 30

(D) Min 4, Q1 7, Median 16, Q3 25, Max 31

2 **Write the name of the correct group on each line.**

Each card describes a part of the stadium box plot. Sort each card under the matching box plot part.

Groups: **Part of the five-number summary** | **Measure of spread**

_____ The minimum — the lowest score

_____ Q1 — the median of the lower half

_____ The median — the middle score

_____ Q3 — the median of the upper half

_____ The maximum — the highest score

_____ IQR — Q3 minus Q1

_____ Range — maximum minus minimum

PART 2 Apply it

real-world problems

3 **Circle the letter of the best answer. Show how you know.**

Two basketball teams' scoring data are displayed as box plots. Team A: Min=40, Q1=52, Median=60, Q3=68, Max=75. Team B: Min=35, Q1=55, Median=61, Q3=65, Max=80. A fan says Team B is better because they scored 80 points once. Is the fan right that Team B is better because of the 80-point game?

(A) Yes — 80 is the highest score in the data

(B) No — Team A's maximum is higher

(C) Yes — the maximum always decides

(D) No — one high game is a single extreme value, not typical performance

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

The Hawks' points per quarter are 2, 4, 5, 7, 8, 10, 12. Resting on the scoreboard, what are the minimum, median, and maximum?

(A) Minimum 0, median 7, maximum 12

(B) Minimum 2, median 7, maximum 12

(C) Minimum 2, median 5, maximum 12

(D) Minimum 2, median 8, maximum 10

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

The Comets scored 6, 10, 14, 18, 22, 26, 30, 34 across 8 games. After ordering, what are Q1 and Q3?

(A) Q1 = 12, Q3 = 28

(B) Q1 = 10, Q3 = 30

(C) Q1 = 14, Q3 = 26

(D) Q1 = 20, Q3 = 34

SHOW YOUR WORK

6 **Circle the letter of the best answer. Show how you know.**

A box plot for the Rockets shows Min = 8, Q1 = 15, Median = 22, Q3 = 30, Max = 38. The middle 50% of their scores falls between which two values?

(A) Between 8 and 38

(B) Between 8 and 22

(C) Between 15 and 30

(D) Between 22 and 38

SHOW YOUR WORK

EXPLAIN YOUR THINKING

The IQR is Q3 − Q1 = 26 − 10 = 16. What does this tell you about the middle 50% of scorers?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

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